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Oxidized Sodium Alginate Regulates Water Activity and Zn^{2+} Transport in Polyacrylamide Hydrogel Electrolytes for Stable Aqueous Zinc-Ion Batteries

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Abstract

The practical operation of aqueous zinc-ion batteries is often limited by insufficient ion transport, water-induced interfacial reactions, and uneven Zn deposition. To address these issues, oxidized sodium alginate (OSA) was introduced into a polyacrylamide (PAM) matrix to form a functional hydrogel electrolyte. At an OSA loading of 0.9 g per batch, the resulting PAM OSA-0.9 hydrogel developed a more open porous structure and showed improved wettability. Compared with the unmodified PAM electrolyte, its ionic conductivity increased from 16.2 to 28.6 mS cm^{-1}, and its apparent Zn^{2+} transference number increased from 0.88 to 0.93. The electrochemical stability window was also extended from 2.47 to 2.64 V. Meanwhile, hydrogen evolution and Zn corrosion were alleviated, and the Zn nucleation and deposition processes became more uniform. As a result, Zn//Zn symmetric cells operated continuously for 1300 h at 1.0 mA cm^{-2} and 1.0 mAh cm^{-2}. In Zn//$NaV_3O_8 \cdot xH_2O$ full cells, 79% of the capacity was retained after 220 cycles at 1.0 A g^{-1}, while 58% remained after 3500 cycles at 5.0 A g^{-1}. These findings show that OSA can simultaneously improve ionic transport and interfacial stability in PAM-based hydrogel electrolytes, thereby supporting durable aqueous Zn batteries.

1. Introduction

The increasing penetration of renewable energy has created a growing need for energy-storage systems that are safe, economical, and suitable for large-scale deployment [1]. Although lithium-ion batteries dominate portable electronics and electric vehicles, their broader use in stationary storage is affected by the safety risks of flammable organic electrolytes, geographically concentrated lithium resources, and relatively high production costs [2]. Aqueous zinc-ion batteries (AZIBs) offer an attractive alternative because they operate with nonflammable aqueous electrolytes. In addition, metallic Zn has a theoretical capacity of 820 mAh g^{-1} and a standard redox potential of − 0.76 V versus the standard hydrogen electrode, together with advantages in resource availability, cost, and environmental acceptability [3–6]. Nevertheless, the conventional $ZnSO_4$-based liquid electrolytes used in AZIBs do not provide sufficient control over the Zn/electrolyte interface during repeated cycling. Spatial variations in the electric field and Zn^{2+} concentration produce an uneven ionic flux, leading to localized Zn growth and dendrite formation. At the same time, electrochemically active water promotes hydrogen evolution and Zn corrosion. These processes consume Zn and electrolyte, disturb the interfacial pH, and favor the accumulation of poorly conductive products, including basic zinc sulfate, thereby lowering Coulombic efficiency and shortening cycle life [7–10]. Considerable efforts have therefore been devoted to concentrated electrolytes, functional additives, artificial protective layers, and modified separators. Although these strategies can alter Zn^{2+} solvation or protect the Zn surface, they often involve trade-offs such as higher cost, more complex fabrication, poorer interfacial contact, or increased resistance to ion migration. Achieving fast ionic transport, mechanical integrity, and sustained interfacial regulation within a single electrolyte system thus remains an important challenge for long-life AZIBs [11–15].

Hydrogel electrolytes combine a water-containing ionic phase with a mechanically supported polymer framework, offering a practical intermediate between liquid and solid polymer electrolytes. The aqueous

domains maintain continuous pathways for Zn^{2+} migration, whereas the hydrophilic network confines the electrolyte, limits excessive water motion, and preserves conformal contact with the electrode surface. This coupled structure can reduce leakage and improve dimensional stability without sacrificing the relatively rapid ion transport characteristic of aqueous systems. It may also moderate local Zn^{2+} concentration gradients and support more even ion delivery at the Zn interface. Consequently, hydrogel performance is strongly governed by network connectivity, pore morphology, and the chemical environment provided by the polymer chains [16–18].

Polyacrylamide (PAM) is widely used as a hydrogel matrix for aqueous Zn batteries because it is readily polymerized, highly hydrophilic, mechanically compliant, and compatible with Zn-salt solutions. Nevertheless, an unmodified PAM network offers limited chemical diversity for simultaneously controlling mobile water and Zn^{2+} migration. Although its amide groups interact with both species, these interactions alone may be insufficient to maintain a stable interfacial environment over extended cycling [19]. One strategy is to enrich the network with additional polar or ionic sites. Oxygen- and phosphorus-containing moieties, as well as charged groups, can modify polymer–water and polymer–ion interactions, thereby influencing ionic conduction and interfacial reaction kinetics. Abbasi et al., for instance, incorporated phosphonated graphene oxide into PAM and obtained a hydrogel electrolyte with higher ionic conductivity than the pristine PAM system [20]. These findings suggest that coupling PAM with hydrophilic natural polymers bearing multiple polar functionalities may provide a simple route to regulate Zn^{2+} transport and improve the reversibility of Zn plating/stripping [21–24].

Oxidized sodium alginate (OSA) is obtained by periodate-mediated oxidation of sodium alginate, a widely available polysaccharide derived from renewable resources. This reaction partially converts vicinal hydroxyl groups into aldehyde functionalities while preserving hydroxyl and carboxylate sites along the alginate chains. When introduced into a hydrogel matrix, these oxygen-containing groups establish additional hydrogen-bonding sites with water molecules and may alter the local aqueous environment, thereby limiting excessive water mobility and reducing water-driven interfacial reactions [25, 26]. The carboxylate groups can also interact electrostatically and coordinatively with Zn^{2+}, whereas hydroxyl and aldehyde groups may contribute to the surrounding hydrogen-bonding and coordination environment. Such multifunctional interactions are expected to influence Zn^{2+} migration through the polymer network and its distribution near the Zn surface. Previous studies on alginate-containing gel electrolytes have reported improved ion transport and more stable electrochemical cycling in aqueous Zn batteries [27–33]. On this basis, combining OSA with PAM offers a bio-derived route to tailor both the internal ion-transport environment and the Zn/electrolyte interface.

Building on this rationale, OSA was integrated into a PAM network to develop a composite hydrogel electrolyte for aqueous Zn-ion batteries. Among the formulations examined, PAM OSA-0.9 GEL provided the most balanced electrochemical performance. The addition of OSA altered the hydrogel microstructure and enhanced surface wettability, which was accompanied by an increase in ionic conductivity from 16.2 to 28.6 mS cm^{-1} and in the apparent Zn^{2+} transference number from 0.88 to 0.93. The composite electrolyte also showed an extended electrochemical stability window, reduced

hydrogen-evolution activity, and improved resistance to Zn corrosion. These changes favored a more uniform Zn deposition process and enabled Zn//Zn symmetric cells to cycle for 1300 h at 1.0 mA cm^{-2} and 1.0 mAh cm^{-2}. Zn//Cu cells further delivered an average Coulombic efficiency approaching 100% over 250 cycles. When coupled with a $NaV_3O_8 \cdot xH_2O$ cathode, the corresponding full cell retained 79% of its capacity after 220 cycles at 1.0 A g^{-1} and 58% after 3500 cycles at 5.0 A g^{-1}. This work therefore highlights the potential of OSA-modified PAM hydrogels for coordinating ion transport and interfacial stability in durable aqueous Zn batteries.

2.Experiment

2.1 Materials

Zinc sulfate heptahydrate ($ZnSO_4 \cdot 7H_2O$, analytical reagent grade, AR), sodium alginate (SA, AR), acrylamide (AM, AR), N,N'-methylenebisacrylamide (MBA, chemically pure grade, CP), sodium periodate ($NaIO_4$, AR), potassium chloride (KCl, AR), and vanadium pentoxide (V_2O_5, 99.5%) were purchased from Aladdin. N-Methyl-2-pyrrolidone (NMP) was purchased from Macklin. Poly(vinylidene fluoride) (PVDF) was obtained from Arkema Kynar. Absolute ethanol (AR) was purchased from IJOK Solvents, and ethylene glycol (AR) was supplied by Shanghai Titan Scientific Co., Ltd. Ammonium persulfate (APS, AR) was purchased from Sinopharm Chemical Reagent Co., Ltd. Acetylene black was obtained from Cyber Electrochemical Materials Network. All chemicals were used as received without further purification.

2.2 Preparation of Oxidized Sodium Alginate

Oxidized sodium alginate (OSA) was synthesized through periodate-mediated oxidation of sodium alginate (SA). SA (4.0 g) was first dissolved in 400 mL of deionized water at room temperature. After complete dissolution, 4.0 g of $NaIO_4$ was introduced, and the reaction was allowed to proceed under stirring for 24 h in the absence of light. Ethylene glycol (1.4 mL) was then added to consume the residual periodate, followed by another 30 min of stirring. OSA was precipitated by introducing 1.6 g of KCl and 400 mL of absolute ethanol into the reaction solution. After centrifugation, the collected solid was dissolved again in 400 mL of deionized water and subjected to a second precipitation using the same amounts of KCl and ethanol. The recovered product was subsequently dispersed in absolute ethanol and centrifuged once more to remove residual impurities. The purified OSA was finally obtained by vacuum drying at 60°C for 6 h.

2.3 Preparation of PAM GEL and PAM OSA-x GEL Electrolytes

A series of PAM–OSA hydrogels containing different amounts of OSA were synthesized by free-radical polymerization. For each formulation, 1.5 g of acrylamide (AM) and x g of OSA (x = 0.3, 0.6, or 0.9) were dispersed in 9 mL of deionized water. Subsequently, 1 mL of N,N'-methylenebisacrylamide (MBA) solution at a concentration of 10 mg mL^{-1} was introduced as the crosslinking agent. After the mixture became uniform, it was sonicated for 10 min to eliminate entrapped air. Ammonium persulfate solution

(200 μL, 20 mg mL^{-1}) was then incorporated to initiate polymerization, and the precursor was mixed thoroughly. The resulting solution was cast into a watch glass with a diameter of 20 mm and thermally polymerized at 80°C for 2 h. The obtained hydrogels were equilibrated in 2 M $ZnSO_4$ solution for 2 h before use as electrolytes. These samples were designated PAM OSA-x GEL, where x denotes the mass of OSA added to each precursor batch. The OSA-free control, denoted PAM GEL, was prepared under identical conditions using the same amounts of AM, MBA, APS, and water, but without the addition of OSA.

2.4 Preparation of the NVO Cathode Material and Cathode Electrode

$NaV_3O_8 \cdot xH_2O$ was synthesized using a previously reported chemical-conversion route with minor modifications [13]. V_2O_5 powder (3.0 g) was dispersed in 90 mL of 2 M NaCl solution and maintained under stirring at 25°C for 72 h. The reddish-brown solid formed during the reaction was isolated by centrifugation and rinsed repeatedly with deionized water and ethanol. After vacuum drying at 60°C overnight, the resulting $NaV_3O_8 \cdot xH_2O$ powder was designated as NVO.

For cathode fabrication, NVO, acetylene black, and poly(vinylidene fluoride) (PVDF) were combined at a mass ratio of 7:2:1. PVDF was dissolved in N-methyl-2-pyrrolidone (NMP) to prepare the binder solution, while NVO and acetylene black were premixed by grinding in an agate mortar for 15 min. The binder solution was introduced gradually during continued grinding, and the slurry consistency was adjusted by adding NMP when necessary. The resulting mixture was spread onto carbon-paper current collectors. After partial evaporation of NMP under ambient conditions, the coated substrates were vacuum-dried at 60°C overnight. Circular cathodes with a diameter of 12 mm were then punched from the dried electrodes. The active-material loading was approximately 1.0 mg cm^{-2}.

2.5 Cell Assembly

CR2032 coin cells were assembled under ambient laboratory conditions. For the Zn//NVO configuration, a 0.1-mm-thick Zn foil served as the anode, the as-prepared NVO electrode as the cathode, and the hydrogel as both the electrolyte and separator. The Zn foil and hydrogel were cut into 14-mm-diameter disks. Cell components were stacked from the negative-case side in the following order: Zn foil, hydrogel electrolyte, NVO cathode, stainless-steel spacer, spring, and positive case. The assembled cells were subsequently sealed with a coin-cell crimping machine under an applied pressure of 50–60 MPa.

Zn//Zn symmetric, Zn//Cu asymmetric, and Zn//stainless-steel (Zn//SS) cells were constructed following the same assembly sequence, with the NVO cathode replaced by Zn, Cu, or stainless-steel foil, respectively. All metal foils used in these configurations were 0.1 mm thick and were cut into 14-mm-diameter disks.

2.6 Materials Characterization

Fourier transform infrared (FTIR) spectra were collected on a Nicolet iZ10 spectrometer to identify the chemical functionalities of the samples. Their crystalline structures were examined by X-ray diffraction (XRD) using a Bruker D2 PHASER diffractometer equipped with Cu Kα radiation. The measurements were conducted at 35 kV over a 2θ range of 5–80°, with a scan rate of 10° min^{-1}. Sample morphologies were observed with a Hitachi Regulus 8100 field-emission scanning electron microscope (FESEM).

2.7 Electrochemical Measurements

Electrochemical tests, except for galvanostatic cycling measurements, were carried out on a CHI760E electrochemical workstation. Cyclic voltammetry (CV), linear sweep voltammetry (LSV), and Tafel polarization were conducted with Zn//stainless-steel (Zn//SS) cells. Chronoamperometric (CA) measurements were performed in Zn//Zn symmetric cells to examine the Zn deposition behavior. The impedance and direct-current polarization data required for determining the apparent Zn^{2+} transference number were also obtained from Zn//Zn cells. For these measurements, the electrochemical impedance spectroscopy (EIS) spectra were collected over a frequency range of 0.01 Hz to 100 kHz.

Galvanostatic Zn plating/stripping of the Zn//Zn symmetric cells, Coulombic-efficiency measurements of the Zn//Cu cells, and cycling tests of the Zn//NVO full cells were performed using a LAND CT2001A battery-testing system.

The apparent Zn^{2+} transference number was evaluated by combining direct-current polarization with EIS measurements recorded before and after polarization [34]. The value was calculated using the Bruce–Vincent equation:

$$t_{Zn2+} = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_S R_S)}$$

where I_0 and I_s are the initial and steady-state currents, respectively; R_0 and R_s denote the corresponding interfacial resistances before and after polarization; and ΔV is the applied polarization voltage.

The ionic conductivity of each hydrogel electrolyte was measured in a stainless-steel blocking-electrode configuration [34]. A hydrogel disk of known thickness and area was sandwiched between two stainless-steel electrodes to form an SS//SS cell. During testing, pressure was applied to maintain close interfacial contact. The corresponding EIS spectra were recorded from 100 Hz to 100 kHz, and the ionic conductivity was calculated as:

$$\sigma = L/R_b S$$

where σ is the ionic conductivity (S cm^{-1}), L is the hydrogel thickness (cm), R_b is the bulk resistance (Ω), and S is the effective contact area between the hydrogel and the stainless-steel electrodes (cm^2).

3. Results and discussion

The preparation of the PAM–OSA electrolyte comprises two successive stages: conversion of SA into OSA and formation of the crosslinked PAM matrix. In the oxidation step shown in Fig. 1(a), periodate attacks the vicinal diol groups located at the C2 and C3 positions of the uronate residues. Oxidative cleavage of the C2–C3 bond disrupts the pyranose ring and produces two aldehyde functionalities, yielding OSA. The subsequent network-construction process is presented in Fig. 1(b). Upon heating, ammonium persulfate (APS) generates sulfate radical species that initiate the polymerization of acrylamide (AM). N,N'-Methylenebisacrylamide (MBA), which contains two polymerizable vinyl groups, creates covalent junctions between the growing PAM chains and establishes a three-dimensional network. During gelation, the OSA chains become embedded and physically entangled within this PAM framework. Intermolecular interactions between OSA and PAM further contribute to the integrity of the resulting composite hydrogel [24, 35, 36].

Fourier transform infrared spectroscopy was used to examine the chemical changes induced by periodate oxidation. In the OSA spectrum (Fig. 1(c)), a new band appears at 1733 cm^{-1} and is assigned to the stretching vibration of aldehyde carbonyl groups [37]. The broad O–H band near 3500 cm^{-1} becomes weaker after oxidation, indicating a reduction in hydroxyl content associated with the consumption of vicinal diols. In addition, the absorption feature related to C–O–C and C–O vibrations shifts from about 1100 cm^{-1} in SA to approximately 1080 cm^{-1} in OSA, accompanied by a decrease in intensity. This change suggests that periodate treatment alters the local bonding environment of the alginate backbone, consistent with C2–C3 bond cleavage and ring opening. Taken together, the appearance of the aldehyde band and the accompanying changes in the hydroxyl and C–O regions support the successful oxidation of SA to OSA [35, 36].

The FTIR profiles of AM, PAM GEL, and PAM OSA-0.9 GEL are presented in Fig. 1(d). Several bands observed for AM at 1424, 1350, and 1279 cm^{-1} are no longer distinguishable in the two hydrogel samples. This spectral evolution is associated with the conversion of the monomer during radical polymerization and is consistent with the formation of a crosslinked PAM matrix [24]. After OSA addition, the broad envelope near 3400 cm^{-1} becomes wider and stronger. Because O–H and N–H stretching vibrations overlap in this region, the variation may arise from the additional hydroxyl-containing groups introduced by OSA and changes in the hydrogen-bonding environment among OSA, PAM, and retained water. A further difference is observed in the amide I region: the band located at about 1660 cm^{-1} for PAM GEL shifts to approximately 1666 cm^{-1} for PAM OSA-0.9 GEL and shows an increase in intensity [38]. The position and intensity changes indicate that the local environment of the carbonyl-containing groups is modified after OSA incorporation. Contributions from the oxygen-containing functionalities of OSA may also overlap with the PAM carbonyl absorption in this region. Overall, the coordinated variations in the O–H/N–H and carbonyl-related bands are consistent with the presence of OSA and altered intermolecular interactions in the composite hydrogel.

X-ray diffraction was employed to probe the structural evolution of sodium alginate after periodate treatment. The SA pattern in Fig. 1(e) displays two broad reflections centered at approximately 13.5° and 22.5°, indicating the presence of relatively ordered regions within the alginate chains [35]. Both

reflections are substantially attenuated in OSA. This loss of diffraction intensity suggests that oxidation disturbs the original chain packing and reduces the extent of short-range structural order. Such disruption is reasonably associated with cleavage of the C2–C3 bond and opening of the uronate ring. The XRD results therefore complement the FTIR evidence for oxidation-induced changes in the alginate backbone.

SEM observations of the freeze-dried samples reveal a clear difference in morphology between PAM OSA-0.9 GEL and PAM GEL. PAM OSA-0.9 GEL presents an interconnected porous framework with apparent pore dimensions mainly in the range of 1–2 μm (Fig. 2(a)). By comparison, the freeze-dried PAM GEL exhibits a more compact morphology, and well-defined pores are scarcely observed (Fig. 2(b)). The more open structure retained by the OSA-containing sample after freeze-drying may increase the accessible surface area and facilitate electrolyte uptake. It may also provide less tortuous pathways for ionic migration within the hydrated gel [39].

Water contact-angle measurements reveal a marked difference in the surface wetting behavior of the two hydrogel electrolytes. The apparent contact angle decreases from 65.71° for PAM GEL to 33.24° for PAM OSA-0.9 GEL (Figs. 2(c) and 2(d)), indicating that the OSA-containing surface is more readily wetted by water. This improvement may be associated with the oxygen-containing functionalities of OSA, including hydroxyl, carboxylate, and aldehyde groups, which increase surface polarity and promote interactions with water. The improved wettability is expected to facilitate electrolyte spreading and help establish closer contact between the hydrogel and electrode surface.

The polarization responses and impedance spectra obtained from Zn//Zn cells are shown in Figs. 2(e) and 2(f). Based on the current and resistance values before and after polarization, the apparent Zn^{2+} transference number was calculated to be 0.93 for PAM OSA-0.9 GEL, compared with 0.88 for PAM GEL. Under the present measurement conditions, this increase indicates that Zn^{2+}-related transport accounts for a larger fraction of the total ionic conduction in the OSA-containing electrolyte. The polar functionalities introduced by OSA may modify the local coordination and distribution of Zn^{2+} within the polymer network. Such changes could promote more Zn^{2+}-dominated ion transport, although the measured value should be regarded as an apparent transference number influenced by both bulk conduction and electrode interfacial processes.

The impedance response of the stainless-steel blocking-electrode cells is presented in Fig. 2(g). PAM OSA-0.9 GEL delivered an ionic conductivity of 28.6 mS cm^{-1}, compared with 16.2 mS cm^{-1} for PAM GEL. The higher conductivity may originate from the combined effects of improved electrolyte wettability, the modified hydrogel morphology, and the additional polar functionalities supplied by OSA. These features can facilitate electrolyte uptake and create a more favorable environment for ionic motion through the hydrated polymer matrix. When considered together with the increase in the apparent Zn^{2+} transference number, the conductivity result indicates lower overall transport resistance and more effective Zn^{2+} delivery under the applied conditions. This transport behavior may alleviate interfacial concentration polarization and contribute to a more even Zn^{2+} distribution during deposition [40, 41].

The OSA content was optimized by comparing the CV, LSV, and Tafel polarization responses of PAM OSA-x GELs with different OSA loadings. As shown in Figs. S1–S3, increasing the OSA content progressively improves the electrochemical stability and corrosion resistance of the gel electrolyte and leads to more favorable Zn nucleation behavior. Among the formulations evaluated, PAM OSA-0.9 GEL exhibits the best overall electrochemical performance. This formulation contains 0.9 g of OSA per batch, which is close to the practical solubility limit under the preparation conditions. Further addition of OSA could result in incomplete dissolution and reduce the homogeneity of the precursor solution. Therefore, PAM OSA-0.9 GEL was selected for subsequent structural characterization and electrochemical evaluation.

The electrochemical stability and hydrogen-evolution behavior of PAM GEL and PAM OSA-0.9 GEL were further evaluated by LSV. As shown in Figs. 3(a)–(c), PAM GEL exhibits an electrochemical stability window from − 0.11 to 2.36 V versus Zn/Zn^{2+}, whereas the window of PAM OSA-0.9 GEL extends from − 0.17 to 2.47 V. PAM OSA-0.9 GEL also exhibits a lower hydrogen-evolution current density, indicating suppressed hydrogen evolution at the Zn anode surface (Fig. 3(b)). These results indicate that OSA incorporation enhances the resistance of the hydrogel electrolyte to reductive and oxidative decomposition while suppressing hydrogen evolution.

Tafel polarization was employed to compare the corrosion behavior of Zn in the two hydrogel electrolytes. As presented in Fig. 3(d), PAM OSA-0.9 GEL yields a lower corrosion current density and a positively shifted corrosion potential relative to PAM GEL. The decrease in corrosion current indicates that the interfacial corrosion reaction proceeds at a slower rate in the OSA-containing electrolyte. This behavior may be related to the polar functionalities of OSA, including hydroxyl, carboxylate, and aldehyde groups. Their interactions with water and Zn^{2+} can modify the local chemical environment near the Zn surface and reduce the availability of reactive water. Such interfacial regulation may consequently limit parasitic reactions and improve the corrosion stability of the Zn electrode.

Chronoamperometric measurements were performed at an overpotential of − 150 mV to compare Zn nucleation and growth in the two hydrogel electrolytes. For the PAM GEL-based cell, the current rises rapidly during the initial 40 s and continues to increase afterward, indicating a prolonged expansion of the electrochemically active deposition area (Fig. 3(e)). Such a response is commonly associated with sustained lateral Zn^{2+} diffusion and less controlled surface growth. In contrast, the current recorded with PAM OSA-0.9 GEL increases only at the beginning of polarization, after which its growth rate decreases within approximately 10 s and the signal gradually approaches a plateau. The earlier stabilization suggests that lateral ion redistribution is more effectively limited and that the deposition process reaches a relatively steady state more rapidly. This behavior is consistent with a more homogeneous interfacial Zn^{2+} distribution and reduced localized Zn accumulation.

Cyclic voltammetry was further used to examine the initial Zn nucleation process. At a scan rate of 5 mV s^{-1}, the cathodic nucleation potential shifted from − 110 mV for the PAM GEL-based cell to − 182 mV for the PAM OSA-0.9 GEL-based cell versus Zn/Zn^{2+} (Fig. 3(f)). This negative shift corresponds to an

increase of approximately 72 mV in the apparent nucleation overpotential, suggesting that a greater electrochemical driving force is required to initiate Zn deposition in the OSA-containing electrolyte. Once nucleation begins, the larger overpotential may increase the nucleation rate and generate a greater number of smaller nuclei, thereby reducing preferential growth at isolated sites. This interpretation is consistent with the more uniform Zn deposition morphology and the improved cycling stability observed for PAM OSA-0.9 GEL [42, 43].

Long-term Zn plating/stripping was compared in Zn//Zn symmetric cells at 1.0 mA cm^{-2} with an areal capacity of 1.0 mAh cm^{-2} (Fig. 4(a)). The cell employing PAM OSA-0.9 GEL maintained a stable voltage response for 1300 h, with a polarization of approximately 82 mV. By contrast, the PAM GEL-based cell showed pronounced voltage oscillations after about 470 h, indicating progressive deterioration of the Zn/electrolyte interface. Such instability is consistent with increasingly heterogeneous Zn deposition and the accumulation of localized surface irregularities during repeated cycling.

The rate tolerance of the two hydrogel electrolytes was further examined in Zn//Zn symmetric cells at 4.0 mA cm^{-2} and 1.0 mAh cm^{-2}. Under these more demanding conditions, the PAM OSA-0.9 GEL-based cell displayed a smaller voltage polarization and preserved a relatively stable plating/stripping response for a longer duration than the cell using PAM GEL (Fig. 4(b)). This result indicates that the OSA-containing electrolyte provides more stable interfacial behavior during rapid Zn deposition and dissolution. Its superior response is consistent with reduced transport resistance and improved regulation of the Zn/electrolyte interface.

The response of the Zn//Zn symmetric cells to stepwise changes in current density was examined from 0.5 to 5.0 mA cm^{-2} at a fixed areal capacity of 1.0 mAh cm^{-2} (Fig. 4(c)). Throughout the current sequence, the PAM OSA-0.9 GEL-based cell showed more regular voltage plateaus and smaller polarization than the PAM GEL-based cell. When the current density was restored to its initial value, the voltage polarization nearly returned to the original level, suggesting that limited irreversible interfacial deterioration occurred during the rate test. The OSA-containing electrolyte therefore exhibited better tolerance to variations in Zn plating/stripping rate. When considered together with its higher ionic conductivity and apparent Zn^{2+} transference number, the reduced polarization is consistent with lower ionic-transport resistance and less severe concentration polarization. These characteristics may support more even Zn^{2+} delivery to the electrode surface during repeated plating and stripping [44, 45].

Zn plating/stripping reversibility was examined in Zn//Cu cells at 2.0 mA cm^{-2} and 2.0 mAh cm^{-2}. The cell equipped with PAM OSA-0.9 GEL sustained a Coulombic efficiency near 100% throughout 250 cycles (Fig. 5(a)), indicating limited irreversible Zn consumption under these conditions. In comparison, the efficiency of the PAM GEL-based cell began to deteriorate noticeably after about 150 cycles. The subsequent instability suggests progressive loss of electrochemically active Zn, together with increasingly severe parasitic reactions at the electrode/electrolyte interface.

The evolution of Zn plating/stripping polarization was further examined from the voltage–capacity curves at selected cycles. For the PAM OSA-0.9 GEL-based cell, the profiles remained closely overlapped

during the first 200 cycles, while the voltage hysteresis stayed below 130 mV (Fig. 5(c)). In comparison, the PAM GEL-based cell showed a hysteresis exceeding 140 mV, together with a progressive widening of the voltage gap upon cycling (Fig. 5(b)). These results indicate that OSA incorporation limits the accumulation of polarization and preserves more stable interfacial kinetics during repeated Zn deposition and removal. Combined with the Coulombic-efficiency results, the smaller voltage hysteresis is consistent with reduced ion-transport resistance and a more stable Zn/electrolyte interface [46, 47].

SEM examination of the cycled Zn electrodes reveals distinct surface features for the two hydrogel electrolytes. After cycling with PAM GEL, the Zn surface is dominated by irregular flake-like deposits and agglomerated protrusions, giving rise to a rough and heterogeneous morphology (Fig. 6(a)). These raised regions may intensify local electric-field and current-density variations, favoring continued Zn accumulation at specific sites. In comparison, the electrode paired with PAM OSA-0.9 GEL retains a flatter and denser surface, with fewer pronounced protrusions, dendrite-like features, and corrosion-related cavities. The relatively uniform post-cycling morphology supports the view that the OSA-containing electrolyte regulates interfacial Zn^{2+} delivery and limits localized deposition during repeated plating and stripping.

XRD patterns of the Zn electrodes collected after 50 cycles were analyzed to compare their crystallographic characteristics and surface by-products. Relative to the electrode cycled with PAM GEL, the sample paired with PAM OSA-0.9 GEL displays a stronger Zn(002) reflection and weaker Zn(100) and Zn(101) signals (Fig. 6(b)). This variation in relative peak intensity suggests that Zn deposition is more strongly oriented along the (002) plane in the OSA-containing electrolyte. Such an orientation is consistent with the smoother and denser surface observed by SEM. In addition, the diffraction signals assigned to basic zinc sulfate are less prominent for the PAM OSA-0.9 GEL-based electrode, suggesting reduced accumulation of sulfate-containing interfacial products. The lower level of these deposits may help preserve electrochemically active surface sites and limit interfacial passivation. These post-cycling results are consistent with the lower polarization, higher Coulombic efficiency, and prolonged Zn plating/stripping stability shown in Figs. 4 and 5 [48].

The compatibility of PAM OSA-0.9 GEL with a full-cell configuration was examined using $NaV_3O_8 \cdot xH_2O$ (NVO) as the cathode and Zn metal as the anode. Cells containing PAM GEL were tested in parallel as the control. The diffraction pattern of the synthesized NVO closely matches the reference reflections of $NaV_3O_8 \cdot xH_2O$ (JCPDS No. 37–0176), with no prominent additional peaks observed (Fig. S4). This agreement supports the predominant formation of the intended NVO phase.

The electrochemical reactions of the Zn//NVO full cells were first examined by cyclic voltammetry over 0.3–1.6 V. As displayed in Fig. 7(a), both cells show two pairs of redox peaks with comparable curve shapes, indicating that the principal redox processes of the NVO cathode are preserved after OSA modification of the electrolyte. The PAM OSA-0.9 GEL-based cell, however, produces larger peak currents and a greater enclosed CV area than its PAM GEL counterpart. These differences reflect a stronger electrochemical response and a larger reversible charge contribution under the applied scan conditions.

The improved response is consistent with the lower ionic-transport resistance, higher apparent Zn^{2+} transference number, and enhanced interfacial wettability of PAM OSA-0.9 GEL.

The rate responses of the Zn//NVO cells are compared in Fig. 7(b). At each applied current density, the cell employing PAM OSA-0.9 GEL delivered a higher specific capacity than the PAM GEL-based cell. The performance gap widened as the current increased, indicating that the OSA-containing electrolyte better preserved cathode utilization under demanding charge–discharge conditions. This behavior is consistent with its higher ionic conductivity and lower ion-transport resistance, which may alleviate concentration polarization and facilitate ion delivery to the NVO cathode at high rates.

Long-term cycling of the Zn//NVO cells was examined at 1.0 and 5.0 A g^{-1}. At 1.0 A g^{-1}, the PAM OSA-0.9 GEL-based cell reached a specific capacity of 413 mAh g^{-1} and retained 325 mAh g^{-1} after 220 cycles, corresponding to approximately 79% capacity retention (Fig. 7(c)). Under the same conditions, the cell employing PAM GEL preserved only 54% of its capacity. A larger difference was observed during extended cycling at 5.0 A g^{-1}. After 3500 cycles, 58% of the capacity remained in the PAM OSA-0.9 GEL-based cell, whereas the PAM GEL counterpart retained 26% (Fig. 7(d)). The improved capacity preservation at both current densities indicates that OSA modification helps maintain full-cell operation during prolonged cycling, particularly under high-rate conditions. This behavior is consistent with reduced ionic-transport resistance and a more stable Zn/electrolyte interface, which may limit polarization growth and parasitic reactions over repeated charge–discharge cycles.

The durability of vanadium-based cathodes is strongly influenced by the local aqueous environment. During extended cycling, reactive water at the cathode/electrolyte interface may accelerate corrosion and promote the dissolution of vanadium-containing species, resulting in the loss of active material and the accumulation of poorly reversible products. Water molecules involved in the charge-storage process may also contribute to lattice distortion and irreversible structural changes in NVO [49–51]. In PAM OSA-0.9 GEL, the hydroxyl, carboxylate, and aldehyde functionalities of OSA provide additional hydrogen-bonding sites and may reorganize the surrounding water network. Such interactions could decrease the availability and mobility of highly reactive water without severely hindering Zn^{2+} conduction. The improved full-cell cycling performance is therefore consistent with a less aggressive aqueous environment at the NVO cathode, together with more stable Zn plating/stripping at the anode. The cooperative regulation of ion transport and the two electrode/electrolyte interfaces may account for the enhanced capacity retention and prolonged operation of the Zn//NVO cell.

The performance of PAM OSA-0.9 GEL was benchmarked against recently reported PAM-based hydrogel electrolytes containing polar functional modifiers. As listed in Table S1, its ionic conductivity, Zn//Zn cycling durability, and full-cell behavior are comparable to those achieved by related systems. However, these values should not be interpreted as a strict ranking because the reported studies differ in electrode configuration, current density, areal capacity, cathode chemistry, active-material loading, and operating voltage range. Within these limitations, PAM OSA-0.9 GEL offers a balanced combination of ionic transport and interfacial stability. In addition, OSA introduces multiple oxygen-containing functionalities

through a readily accessible bio-derived component, providing a relatively straightforward route for modifying PAM hydrogel electrolytes.

4. Conclusions

Oxidized sodium alginate was introduced into a polyacrylamide hydrogel as a bio-derived functional component for regulating ionic transport and the Zn/electrolyte interface. The incorporation of OSA modified the hydrogel morphology and surface wettability while providing additional oxygen-containing sites capable of interacting with Zn^{2+} and water. Consequently, the ionic conductivity increased from 16.2 to 28.6 mS cm^{-1}, and the apparent Zn^{2+} transference number rose from 0.88 to 0.93. The OSA-containing electrolyte also showed lower hydrogen-evolution activity and slower Zn corrosion, together with more controlled nucleation and deposition behavior. After cycling, the corresponding Zn electrode retained a smoother and denser surface, displayed a stronger Zn(002) texture, and contained fewer detectable basic zinc sulfate-related products.

Among the investigated formulations, PAM OSA-0.9 GEL delivered the most balanced electrochemical response. Zn//Zn symmetric cells cycled for 1300 h at 1.0 mA cm^{-2} and 1.0 mAh cm^{-2}, while Zn//Cu cells maintained a Coulombic efficiency close to 100% over 250 cycles. In addition, the Zn//NVO full cell retained 58% of its capacity after 3500 cycles at 5.0 A g^{-1}. The sustained full-cell operation is consistent with lower ionic-transport resistance and improved stability of the Zn interface. A modified local aqueous environment may also contribute to reducing parasitic reactions at the NVO cathode, although direct cathode-side characterization is required to verify this contribution. Overall, OSA modification offers a straightforward means of coordinating ion transport and interfacial regulation in PAM-based hydrogel electrolytes for durable aqueous Zn-ion batteries.

Declarations

Credit authorship contribution statement

Mingyang Chen: Writing-review and editing, Writing-original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Xuanying Huo:** Visualization, Data curation. **Gaosheng Tan :** Visualization, Data curation. **Cunxin Wang:** Visualization, Data curation. **Juan Wang:** Writing-review and editing, Resources, Investigation, Funding acquisition. **Qin Zhong:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No datasets were generated or analysed during the current study.

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Figures

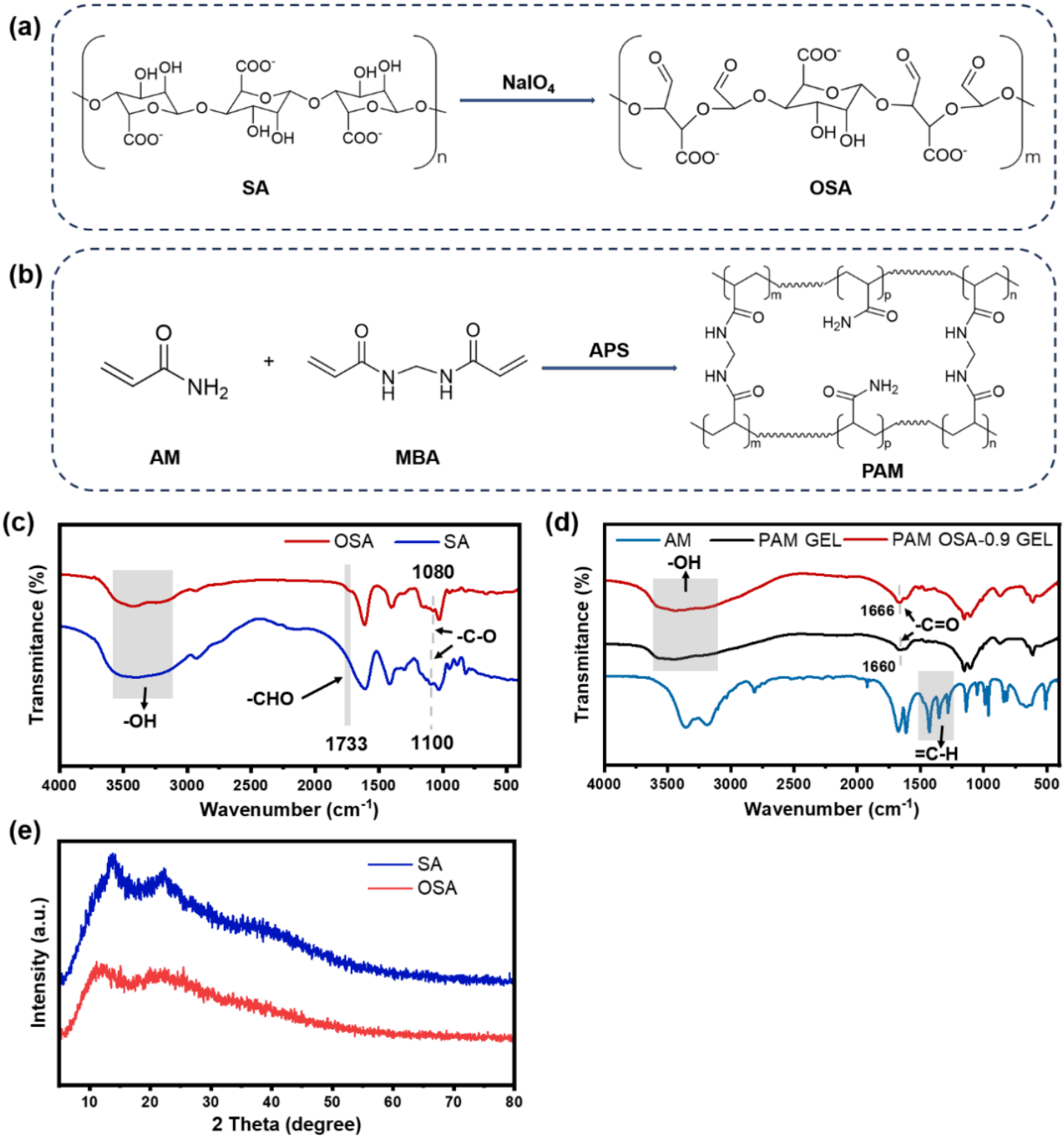


Figure 1

Synthesis and structural analysis of OSA and PAM–OSA hydrogels: (a) periodate-mediated conversion of SA to OSA; (b) APS-initiated formation of an MBA-crosslinked PAM network containing OSA; (c) FTIR spectra of SA and OSA; (d) FTIR spectra of AM, PAM GEL, and PAM OSA-0.9 GEL; (e) XRD patterns of SA and OSA

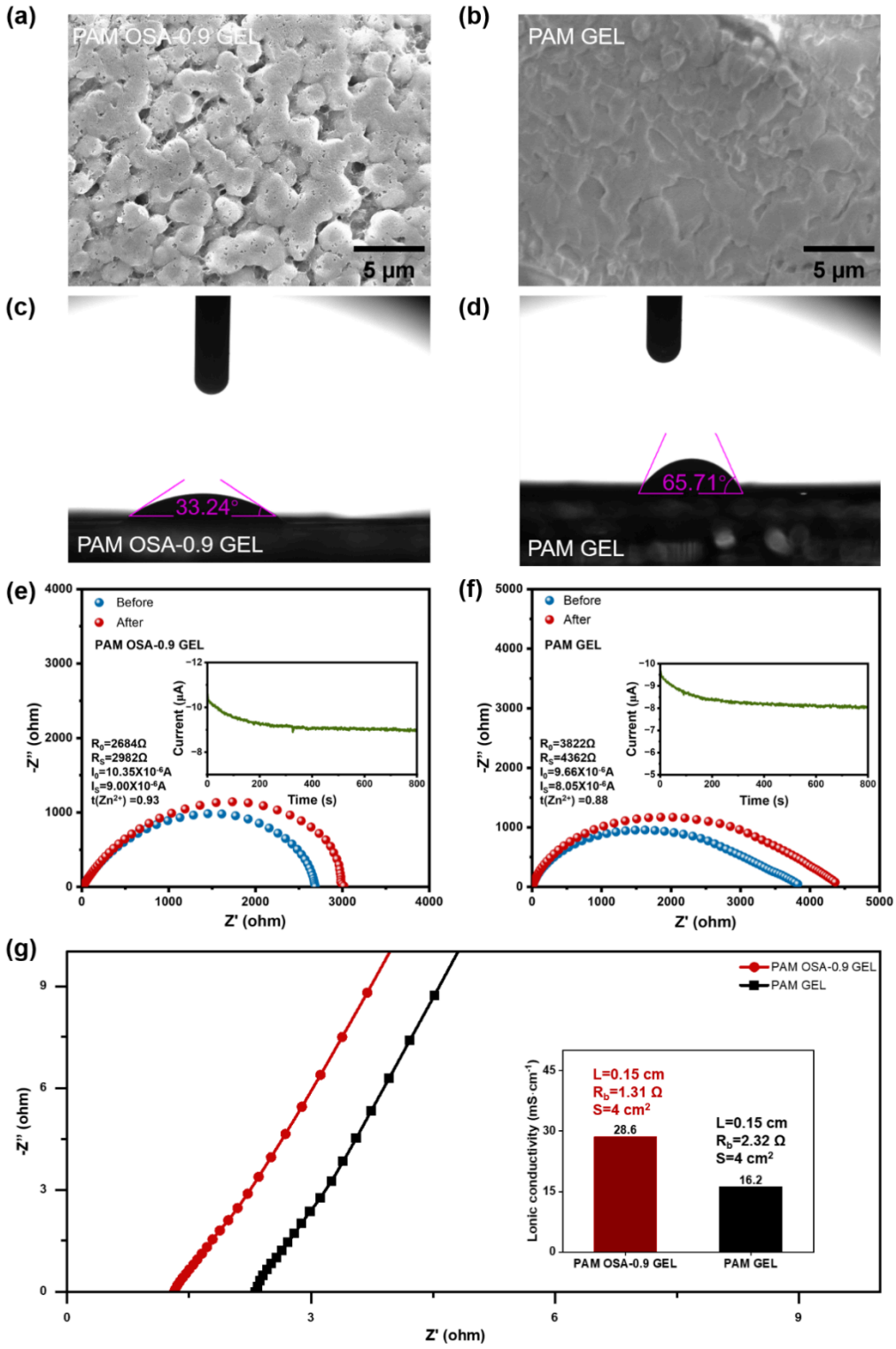


Figure 2

Morphology, surface wettability, and ion-transport behavior of PAM GEL and PAM OSA-0.9 GEL: SEM micrographs of freeze-dried (a) PAM OSA-0.9 GEL and (b) PAM GEL; water contact-angle images of (c) PAM OSA-0.9 GEL and (d) PAM GEL; EIS spectra recorded before and after 10 mV DC polarization for Zn//Zn symmetric cells containing (e) PAM OSA-0.9 GEL and (f) PAM GEL, with the corresponding

current–time curves shown in the insets; (g) EIS spectra of PAM GEL and PAM OSA-0.9 GEL in stainless-steel blocking-electrode cells, with the calculated ionic conductivities shown in the inset

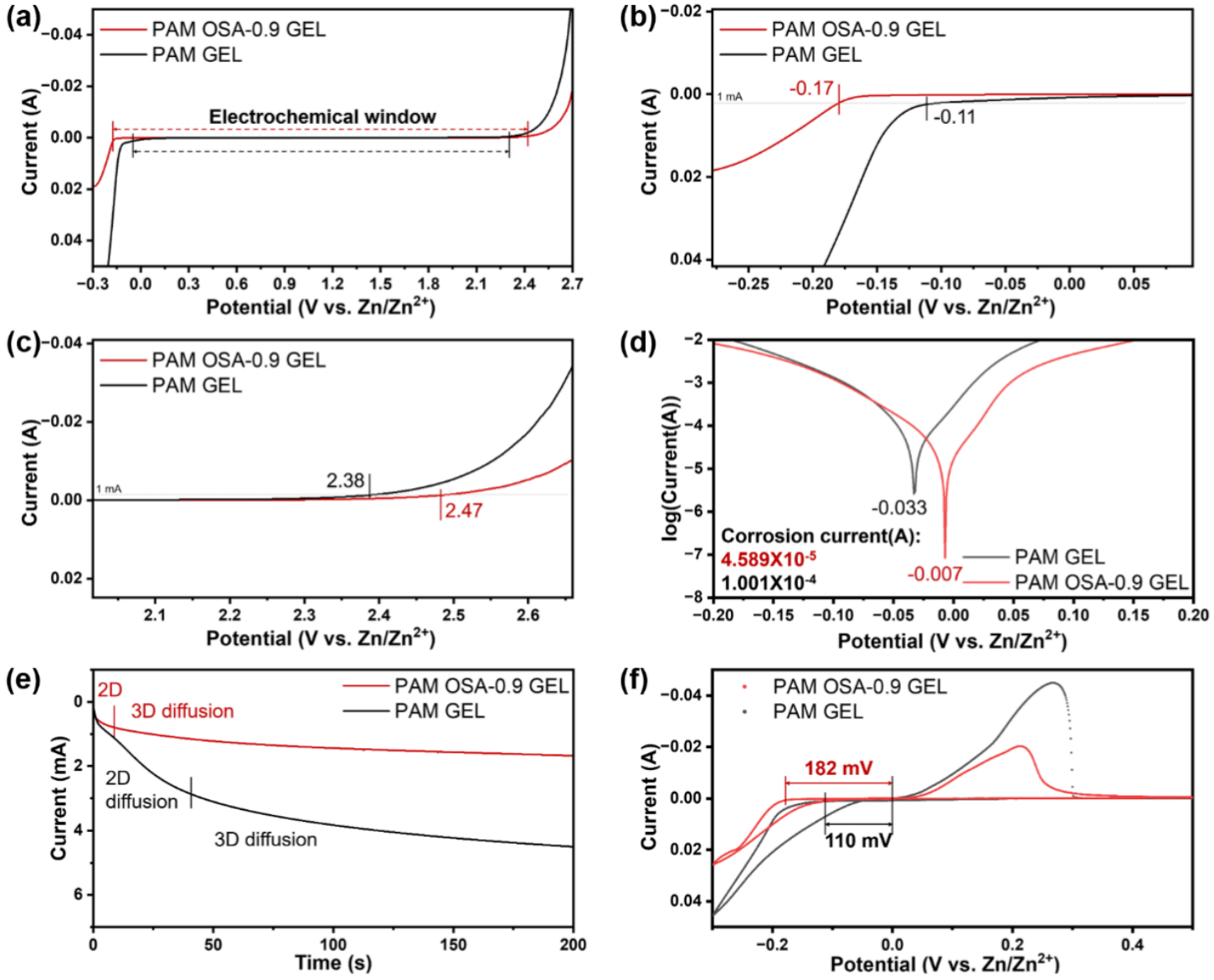


Figure 3

Electrochemical properties of PAM GEL and PAM OSA-0.9 GEL: (a) LSV profiles of different electrolytes; (b) HER and (c) OER profile; (d) Tafel profiles of different electrolytes; (e) CA curves of Zn//Zn cells at a constant overpotential of −150 mV; (f) CV curves of Zn//SS cells at a scan rate of 5 mV

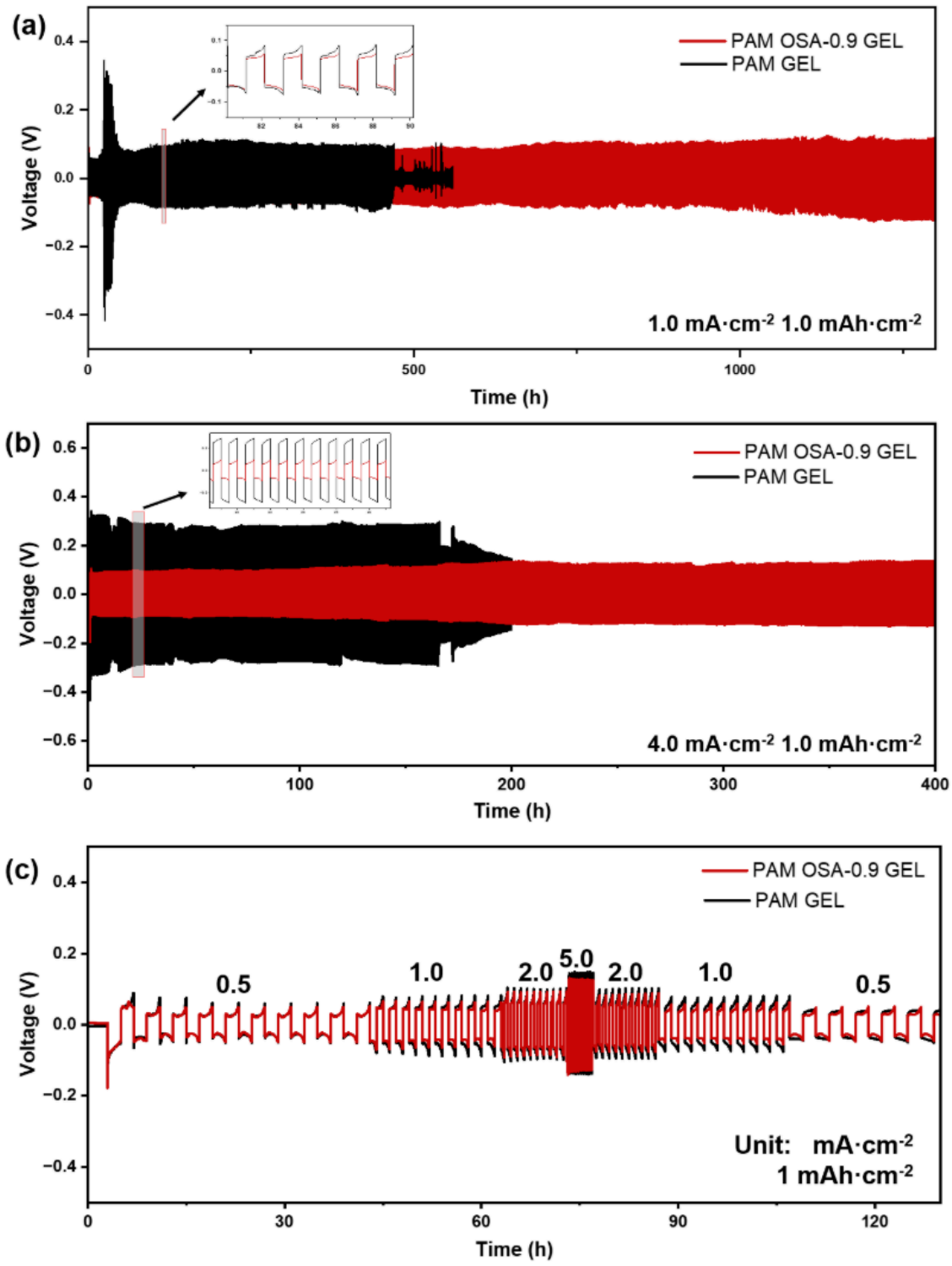


Figure 4

Zn plating/stripping behavior of symmetric cells containing PAM GEL or PAM OSA-0.9 GEL: (a) long-term cycling at 1.0 mA cm^{-2} and 1.0 mAh cm^{-2}; (b) galvanostatic cycling at 4.0 mA cm^{-2} and 1.0 mAh cm^{-2}; (c) voltage response to stepwise current densities from 0.5 to 5.0 mA cm^{-2} at a constant areal capacity of 1.0 mAh cm^{-2}

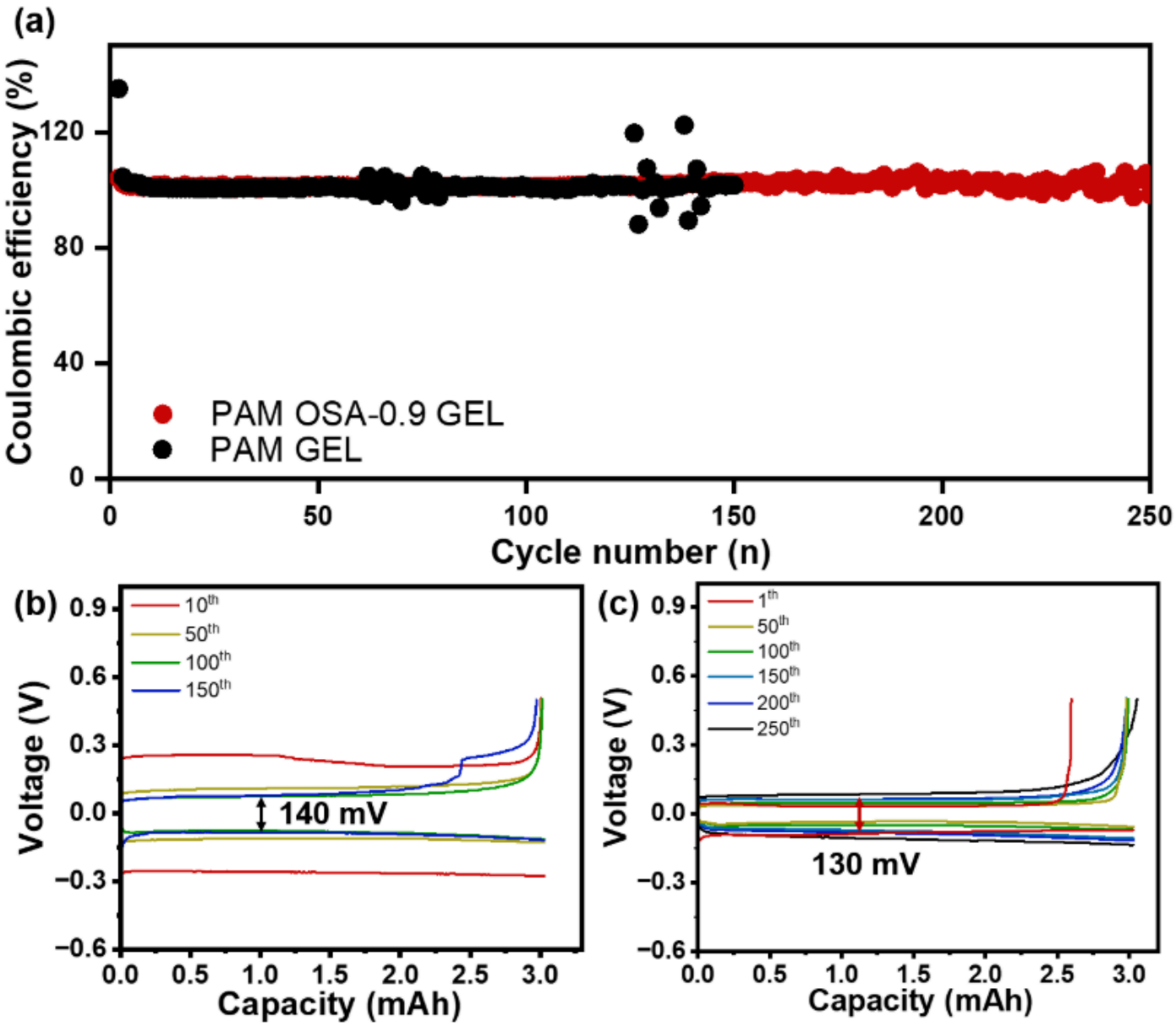


Figure 5

Zn deposition/stripping behavior in Zn//Cu cells containing PAM GEL or PAM OSA-0.9 GEL: (a) Coulombic efficiency during cycling at 2.0 mA cm^{-2} and 2.0 mAh cm^{-2}; representative voltage–capacity curves at selected cycles for cells containing (b) PAM GEL and (c) PAM OSA-0.9 GEL under the same conditions

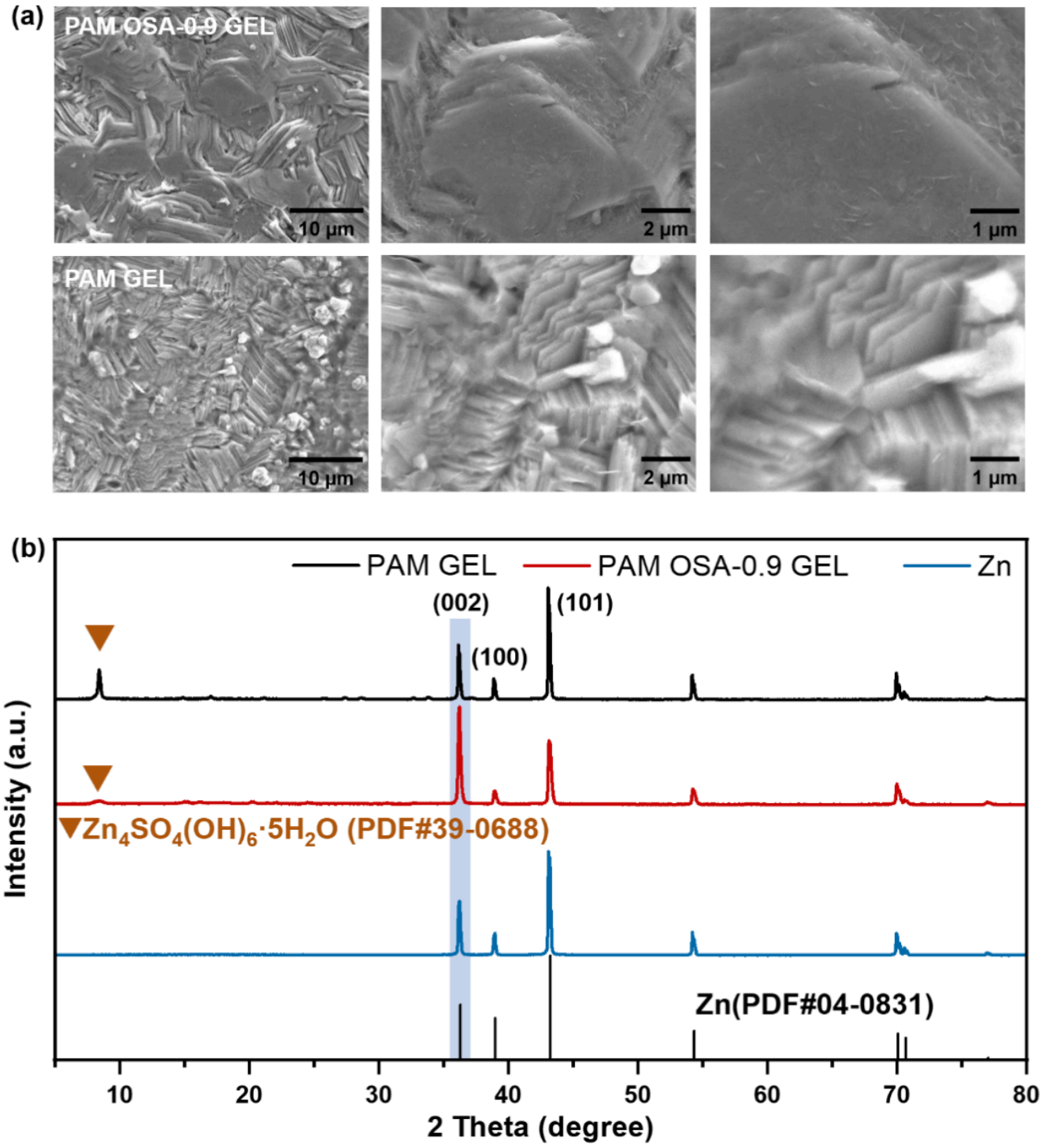


Figure 6

Characterization of zinc anodes cycled with PAM GEL and PAM OSA-0.9 GEL: (a) SEM images of the zinc anodes after cycling; (b) XRD patterns of the pristine zinc foil and the zinc anodes cycled for 50 cycles with PAM GEL and PAM OSA-0.9 GEL

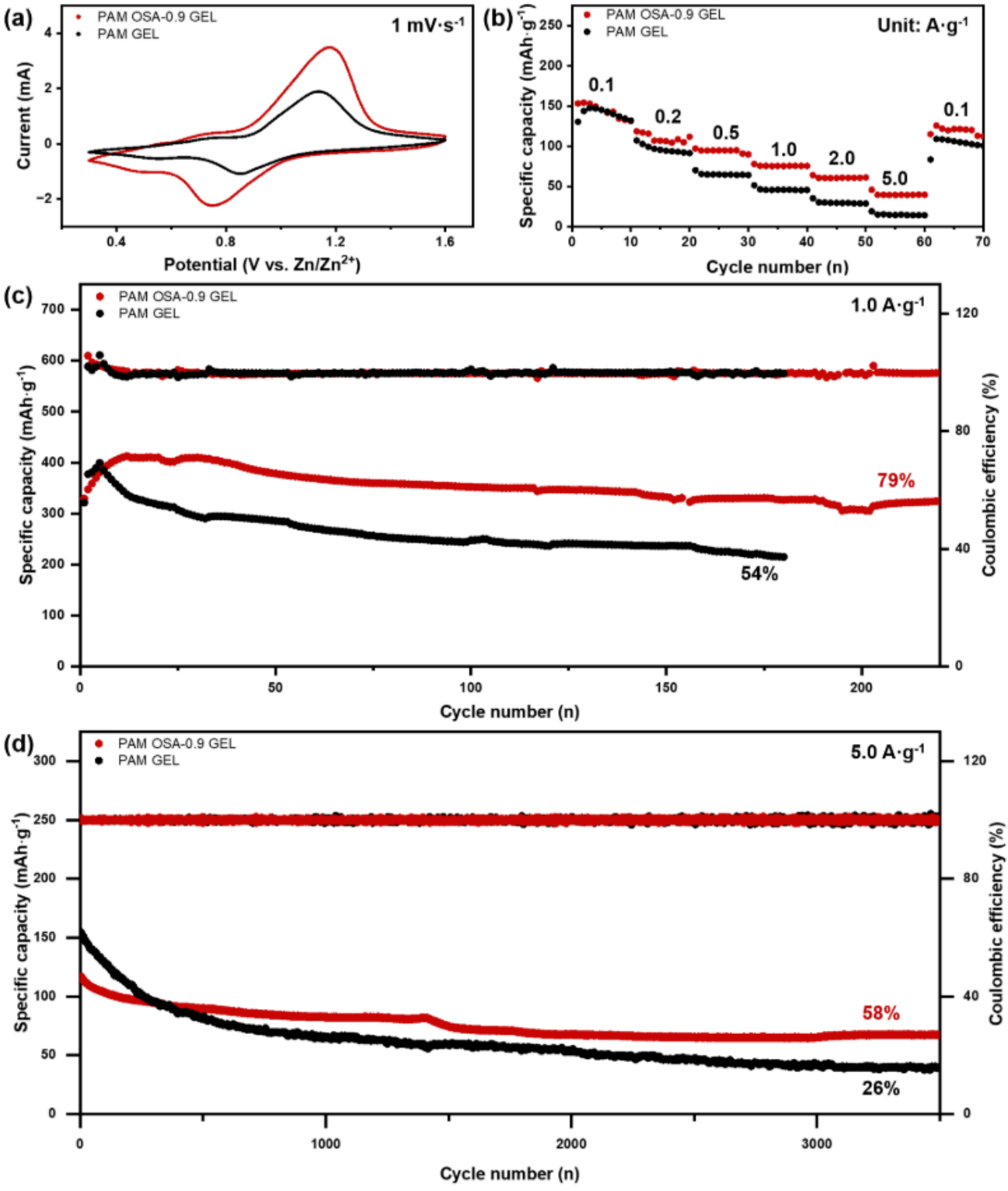


Figure 7

Electrochemical behavior of Zn//NVO full cells containing PAM GEL or PAM OSA-0.9 GEL: (a) cyclic voltammetry profiles; (b) specific capacities at different current densities; long-term cycling at (c) 1.0 A g^{-1} and (d) 5.0 A g^{-1}

Supplementary Files

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